

Exploratory Analyses II

Today's agenda:

- Discuss Assignment 1
- Discuss potential data sources
- Apply chapter 2 techniques to your own data
- Work through 2.6

Research Opportunity

Assignment 1

Full information online (see Canvas/Github)

Assignment 1

Overview:

- Select a dataset (you want one that can ask multiple questions)
- Do some exploratory analyses (and document them)

Assignment 1

In this assignment, students will need to:

1. Select a dataset, provide a description, note where it came from
2. Read in their dataset
3. Note the class of the dataset itself (e.g., matrix, list, dataframe, etc.)
4. List the variables of interest and their classes (e.g., numeric, character, etc.)
5. Examine the structure of the dataset and correct any formatting mistakes
6. Provide summary statistics for variables of interest
7. Produce at least 3 different figures that convey different information
8. For the figures in 6, briefly describe what the figures show

You already have the tools

Step	Where we did it
1. Describe it, cite it	READMEs, Lecture 4
2. Read it in	Lecture 4: <code>read.csv</code> , <code>read_xlsx</code>
3. Class of the dataset	Lecture 4: <code>class()</code>
4. Variables and classes	Lecture 4: <code>str()</code>
5. Fix formatting problems	Lecture 4: <code>-999</code> . Lecture 5: the kiwi
6. Summary statistics	Lecture 4: <code>summary()</code> , <code>table()</code>
7. Three figures	Lecture 5: <code>hist</code> , <code>plot</code> , <code>boxplot</code> , <code>barplot</code>
8. Describe the figures	Lecture 5: what log-log changed

We're going to get more practice with some of these today as well

What counts as three different figures

"Three **different figures** that convey different information"

Three scatterplots of different variables is **one kind** of figure.

Different information means a different question, e.g.:

How is one variable distributed? `hist`

How do two relate? `plot`

How do groups compare? `boxplot`

How many in each category? `barplot`

Assignment 1: what you'll submit

Needs to document what you did and be reproducible

Either:

- 1) The output of an .Rmd file showing code and outputs
 - a) See **example_R_markdown/** for .Rmd file and potential output types, e.g. html, pdf

- 2) A .R file that can be run anywhere
 - a) See **example_R_commented/** for .R file with comment examples

How and when

Assignments can be submitted on canvas.

You can either:

- a) Upload your file to Github and submit the link on Canvas
(require for grads, extra credit undergrads)
- b) Upload the file on canvas

Due Friday, Sept. 18th before midnight.

"Runs anywhere" includes the data

```
# Only works on my laptop
```

```
read.csv("C:/Users/bmaitner/Desktop/.../AVONET1_BirdLife.csv")
```

```
# Works if you have the course repo open
```

```
read.csv("data/Avonet/AVONET1_BirdLife.csv")
```

```
# Works for anyone, anywhere
```

```
read.csv("https://github.com/bmaitner/biometry\_course/raw/refs/heads/main/data/Avonet/AVONET1\_BirdLife.csv")
```

Using your own data? Put it in your repo and link it, or link the original source.

Documenting what you did

Open `example_R_commented/Example.R`

Comments say why, not what:

```
x <- log10(mass)    # take the log of mass (not great)
```

```
x <- log10(mass)    # mass spans 5 orders of magnitude (better)
```

Ctrl(cmd) + shift + r makes a section

Ctrl(cmd) + shift + c makes a comment (or comments out highlighted code)

Ctrl(cmd) + shift + o opens an outline

Try these command out

Questions so far?

Getting data

- Collect it
 - pro: you know it best, can be tailored to a specific question
 - con: time-consuming, possibly expensive, probably requires cleaning
- Download it
 - pro: quick
 - con: less knowledge of collection, probably still needs cleaning
- Aggregate it
 - pro: data volume and breadth
 - con: even less knowledge of collection, lots of standardization and cleaning required

Where to look for data

Course Github site:

- Links to resources
- Data folder contains some options

Scientific Publications

- Supplementary Information
- Data Publications (e.g., <https://www.nature.com/sdata/>)

Data repositories

- Dryad <https://datadryad.org/>
- TRY <https://www.try-db.org/TryWeb/Home.php>

Datasets of interest to this class

- Mangrove invert communities
- Marine microorganisms
- Deep sea megabenthos
- Bird Survey Data
- Coral populations
- Reef fish data

PNAS RESEARCH ARTICLE ECOLOGY BIOLOGICAL SCIENCES OPEN ACCESS

A functional analysis reveals extremely low redundancy in global mangrove invertebrate fauna

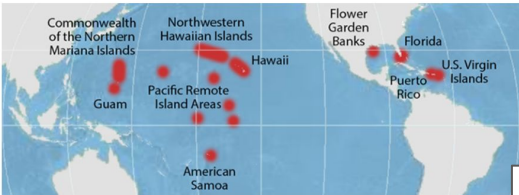
Stefano Cannicci^{1,a,b}, Shing Yip Lee^{2,c,d,1}, Henrique Bravo^{3,a,e}, Jaime Ricardo Cantera-Kintz^{4,f}, Farid Dahdouh-Guebas^{5,g,h}, Sara Fratini^{6,b}, Marco Fusi^{7,i,j}, Pedro J. Jimenez^{8,a}, Inga Nordhaus^{9,k,l}, Francesca Porri^{10,m,n}, and Karen Diele¹¹

Edited by Hugh P. Possingham, The Nature Conservancy, Sherwood, QLD, Australia, and approved May 11, 2021 (received for review August 20, 2020)

July 26, 2021 | 118 (32) e2016913118 | <https://doi.org/10.1073/pnas.2016913118>

NOAA's National Coral Reef Monitoring Program

Coral reefs are among the most valuable ecosystems on earth, providing people with goods and services that include food, storm protection, and recreational opportunities. Despite their importance, coral reef ecosystems are in decline from a myriad of man-made and natural threats.



www.nature.com/scientificdata

SCIENTIFIC DATA

OPEN

SUBJECT CATEGORIES

- » Biodiversity
- » Ecological genetics
- » Ocean sciences
- » Biogeography

Open science resources for the discovery and analysis of Tara Oceans data

Stéphane Pesant^{1,2}, Fabrice Not^{3,4}, Marc Picheral^{5,6}, Stéphanie Kandels-Lewis^{7,8}, Noan Le Bescot⁹, Gabriel Gorsky^{5,6}, Daniele Iudicone⁹, Eric Karsenti^{10,11}, Sabrina Speich^{12,13}, Romain Troublé¹³, Céline Dimier⁷, Sarah Seaton^{14,15} & Tara Oceans Consortium Coordinators*

The North American Breeding Bird Survey, Analysis Results 1966 - 2019

View

Dates

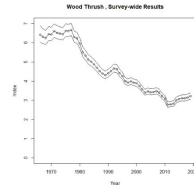
Publication Date : 2020-11-03
Start Date : 1966
End Date : 2019

Citation

Sauer, J.R., Link, W.A., and Hines, J.E., 2020. The North American Breeding Bird Survey, Analysis Results 1966 - 2019: U.S. Geological Survey data release, <https://doi.org/10.5066/966P96A7675>.

Summary

This data product consists of a database of population change and abundance estimates for North American birds, estimated from North American Breeding Bird Survey (BBS) data. Data are presented for 549 species of birds in 4 spreadsheets containing trend estimates and annual indices for 2 time periods. Estimates are derived for each species using the 1 of 4 alternative models, and a cross-validation model selection procedure was used to select the best model for each species. Metadata for the BBS data used to produce these estimates is available from a USGS ftp site (<ftp://ftp.usgs.gov/birds/nabbs/BBSDataFiles/>). Metadata associated with this data product provides information specific to the analysis results.



Example of Annual Indices from Breeding Bird Survey Analysis

Keogh, Poppy (2022): Megabenthic composition and associated environmental variables of the Charlie-Gibbs Fracture Zone [dataset]. PANGAEA, <https://doi.org/10.1594/PANGAEA.946727>

Always quote citation above when using data! You can download the citation in several formats below.

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RIS Citation BibTeX Citation Copy Citation Share Show Map Google Earth 21 8 3

DRYAD

Dynamic riskscapes for prey: Disentangling the impact of human and cougar presence on deer behavior using GPS smartphone locations

Abernathy, Heather¹; Diltner, Mark²; Stoner, David³; Hersey, Kent⁴; Schoenecker, Kathryn⁵; Jackson, Patrick⁶; Engebretsen, Kristin³; Young, Julie³; Wittmyer, George²

Datasets of interest to this class

- General places to look for data noted on the course Github homepage
- Data folders on course Github
 - The “README” files in each folder contain useful information
 - May need to look up the data sources to understand all the column names.

Now do it to your own data

Open a new script. Read your dataset in.

I'll show each step on the bird data. Then you do it on yours.

By the end you'll have three figures, labelled and saved.

That's Assignment 1, steps 6 to 8.

No dataset yet? Use AVONET for now, modify your code later.

Assessing many variables at once

Note: if using a different dataset, use your variables

```
beak <- avonet[c("Beak.Length_Culmen",  
                 "Beak.Length_Nares",  
                 "Beak.Width",  
                 "Beak.Depth")]
```

```
round(cor(beak), 2)
```

What do these lines of code do?

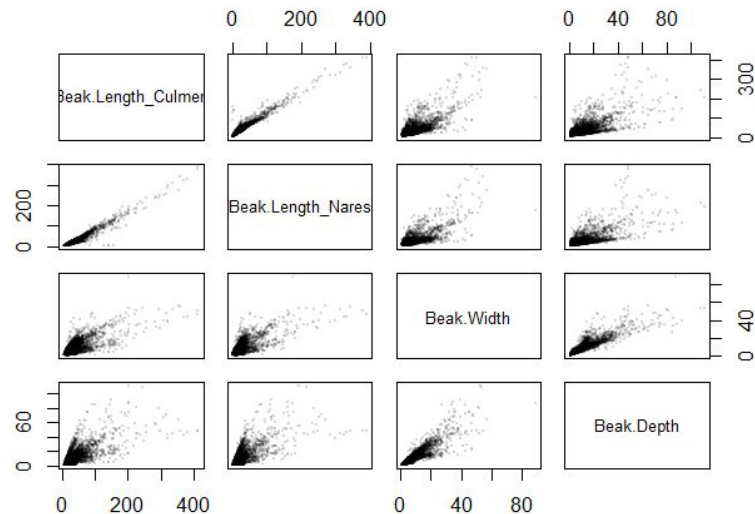
	Culmen	Nares	Width	Depth
Beak.Length_Culmen	1.00	0.97	0.74	0.71
Beak.Length_Nares	0.97	1.00	0.73	0.70
Beak.Width	0.74	0.73	1.00	0.91
Beak.Depth	0.71	0.70	0.91	1.00

The same thing as a picture

```
pairs(x = beak,  
      pch = 16,  
      cex = 0.3,  
      col = rgb(red = 0,  
                 green = 0,  
                 blue = 0,  
                 alpha = 0.15)  
      )
```

Same code, different style:

```
pairs(beak, pch=16, cex=0.3, col=rgb(0,0,0,0.15)  
      )
```



Try this on your data

Correlations in your own data

Did anyone find any suspiciously high correlations?

Or anything that might need log-scaling?

Comparing groups

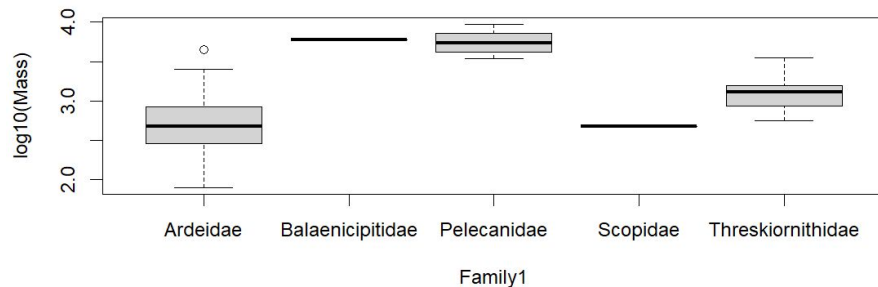
```
pelicans <- avonet[which(  
  avonet$Order1 == "Pelecaniformes", ,
```

```
nrow(pelicans)      # 110
```

```
boxplot(log10(Mass) ~ Family1,  
  data = pelicans)
```

Note the use of `which()` to select only Pelecaniformes!

Try a boxplot on your data (figure 1 of 3!)



Two plots side by side

```
carnivores <- avonet[which(  
  avonet$Trophic.Level == "Carnivore"), ]
```

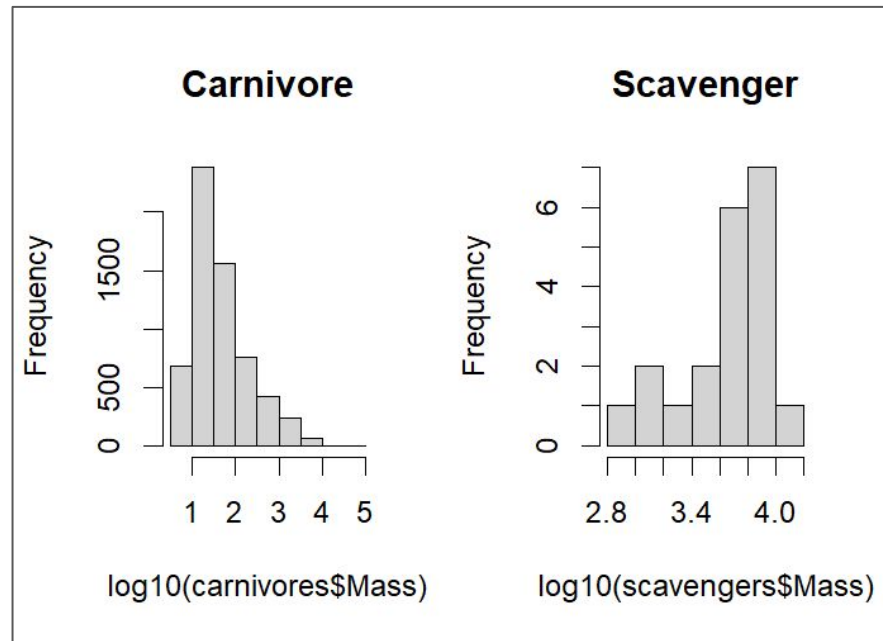
```
scavengers <- avonet[which(  
  avonet$Trophic.Level == "Scavenger"), ]
```

```
par(mfrow = c(1, 2))
```

```
hist(log10(carnivores$Mass),  
     main = "Carnivore")
```

```
hist(log10(scavengers$Mass),  
     main = "Scavenger")
```

```
par(mfrow = c(1, 1)) # put it back!
```



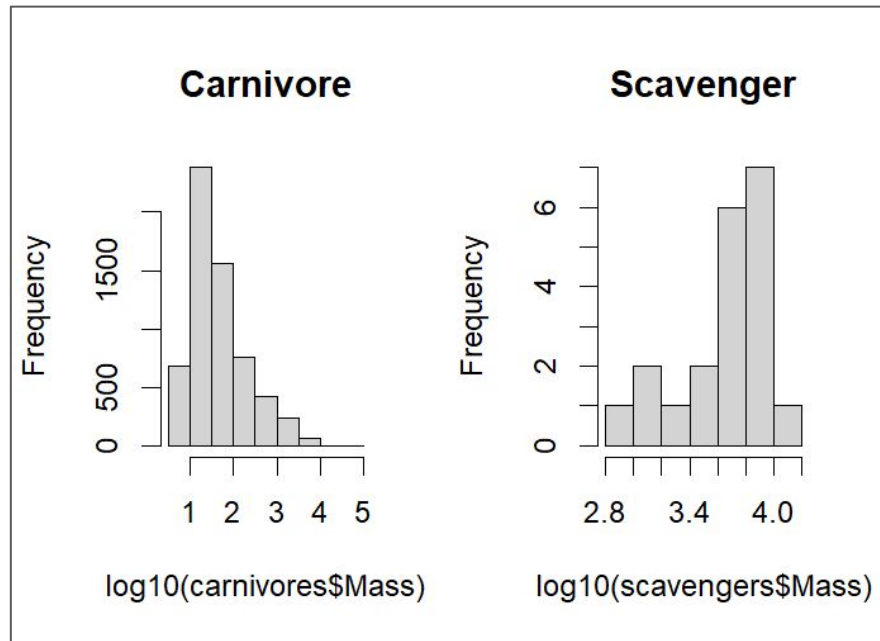
Look before you believe it

Scavengers look about as common as carnivores.

Read the y axes.

We can sanity check with:

```
nrow(carnivores) # 6115  
nrow(scavengers) # 20
```



Two plots side by side

```
carnivores <- avonet[which(  
  avonet$Trophic.Level == "Carnivore"), ]
```

```
scavengers <- avonet[which(  
  avonet$Trophic.Level == "Scavenger"), ]
```

```
par(mfrow = c(1, 2))
```

```
hist(log10(carnivores$Mass),  
     main = "Carnivore")
```

```
hist(log10(scavengers$Mass),  
     main = "Scavenger")
```

```
par(mfrow = c(1, 1))  # put it back!
```

Your turn.

Plot two groups from your data
side by side

Potential figure 2 of 3.

Four groups, four blocks

What about all four groups?

```
herbivores <- avonet[which(  
  avonet$Trophic.Level == "Herbivore"), ]  
omnivores <- avonet[which(  
  avonet$Trophic.Level == "Omnivore"), ]
```

...then four `hist()` calls, differing by one word each, with `ylim` written out four times.

What's the issue with this approach?

For loops

A `for()` loop does it once

```
for (level in c("Carnivore", "Herbivore",  
               "Omnivore", "Scavenger")) {  
  print(level)  
}
```

Runs the code in `{ }` once per element, setting `level` to each in turn.

Now we put the plotting code inside!

For loops

```
par(mfrow = c(2, 2))
for (level in c("Carnivore", "Herbivore",
                "Omnivore", "Scavenger")) {
  print(level)
  group_data <- avonet[which(
    avonet$Trophic.Level == level), ]

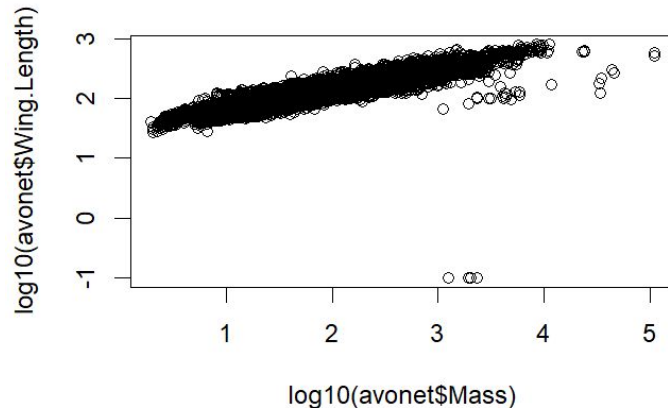
  hist(log10(group_data$Mass),
        main = level,
        xlab = "log10(Mass)",
        ylim = c(0, 4000), breaks = c(0:6))
}
par(mfrow = c(1, 1))
```

Making figures readable

```
plot(x = log10(avonet$Mass),  
     y = log10(avonet$Wing.Length))
```

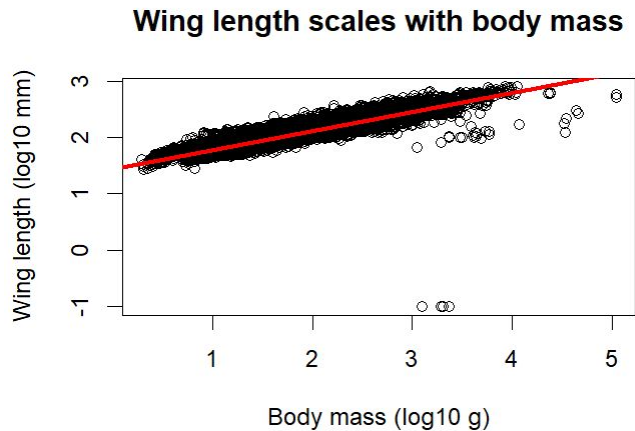
Not bad, but could be improved.

How can we see the available arguments for plot()?



Making figures readable

```
plot(x = log10(avonet$Mass),  
     y = log10(avonet$Wing.Length),  
     xlab = "Body mass (log10 g)",  
     ylab = "Wing length (log10 mm)",  
     main = "Wing length scales with body mass")
```

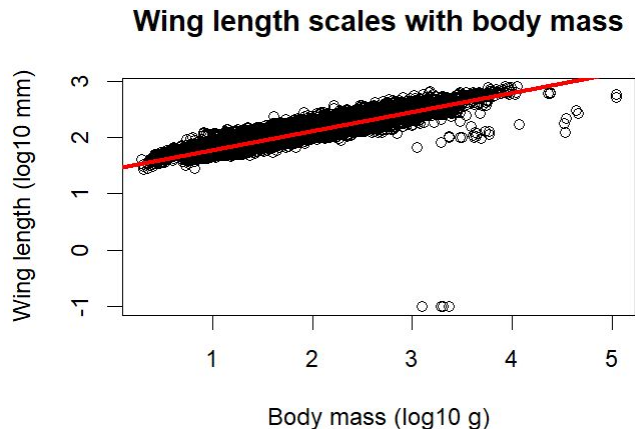


```
fit <- lm(log10(Wing.Length) ~ log10(Mass),  
          data = avonet)
```

```
abline(fit, col = "red", lwd = 3)
```

Making figures readable

```
plot(x = log10(avonet$Mass),  
     y = log10(avonet$Wing.Length),  
     xlab = "Body mass (log10 g)",  
     ylab = "Wing length (log10 mm)",  
     main = "Wing length scales with body mass")
```



```
fit <- lm(log10(Wing.Length) ~ log10(Mass),  
          data = avonet)
```

```
abline(fit, col = "red", lwd = 3)
```

Your turn! Make a scatterplot from your own data. (potential plot 3)

Saving a figure

```
png("my_figure.png",  
    width = 900, height = 650)  
plot(log10(avonet$Mass),  
      log10(avonet$Wing.Length))  
dev.off()    # nothing is written until this
```

Nothing appears in the Plots pane. That's normal.

Your turn. Save all three figures.

Write a commented-out sentence below the code describing what the plot shows.

Remainder of class:

- Continue working on 2.6
- Start working on Assignment 1

Next class:

- Read and work through R for Data Science Chapter 3:
<https://r4ds.hadley.nz/data-transform.html>